

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0994

Monfort 230 kV
April 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0994. The Interconnection Customer proposes a 830.9 MW natural gas combined cycle facility interconnecting at the Monfort 230 kV substation in the Indiana area. The request was received on May 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 286 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,731,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The AEP PETTIT AVENUE to WAYNEDALE SUBSTATION 100.0 kV #1 is overloaded to 129.0% of its 220 MVA emergency rating for the loss of the MUSKOGEE to MONTPELIER ELECTRIC GENERATING STA 500.0 kV Line. The project contributes 5.3% DFAX (43.7 MW) to this overload.

The WAYNEDALE to AEP PETTIT AVENUE 100.0 kV is overloaded to 129.0% of its 220 MVA emergency rating for the loss of the MUSKOGEE to MONTPELIER ELECTRIC GENERATING STA 500.0 kV L. The project contributes 5.3% DFAX (43.7 MW) to this overload.

INDEGO to WAYNEDALE 100.0 kV L: 121.4% loading (134 MVA) following the outage of the MONFORT to WAYNEDALE 230.0 kV line; DFAX = 13.5%.

The WABASH to MONFORT SUBSTATION 230.0 kV is overloaded to 110.5% of its 1,056 MVA emergency rating for the loss of the RICHLAND to DEKALB 500.0 kV Line. The project contributes 47.9% DFAX (397.6 MW) to this overload.

The INDEGO to COLUMBIA HILLS 100.0 kV is overloaded to 108.3% of its 194 MVA emergency rating for the loss of the MONFORT to WAYNEDALE 230.0 kV. The project contributes 13.3% DFAX (110.8 MW) to this overload.

PEAVINE to HOHENWALD 100.0 kV Line: 106.6% loading (120 MVA) following the outage of the WAYNEDALE to STATE STREET 230.0 kV; DFAX = 7.1%.

The MONFORT to NE HARRIMAN 100.0 kV is overloaded to 106.5% of its 134 MVA emergency rating for the loss of the WABASH to MONFORT 230.0 kV. The project contributes 8.9% DFAX (74.0 MW) to this overload.

The PONNEQUIN SUBSTATION to DEER CREEK 100.0 kV Line is overloaded to 105.9% of its 152 MVA emergency rating for the loss of the WABASH 500 kV/230 kV Auto. The project contributes 7.8% DFAX (65.2 MW) to this overload.

The EASTOWN ROAD to DEAN FOREST 230.0 kV is overloaded to 105.4% of its 846 MVA emergency rating for the loss of the MUSKOGEE to SWAN VLY 500.0 kV. The project contributes 9.3% DFAX (77.6 MW) to this overload.

For the loss of SWAN VLY to MUSKOGEE 500.0 kV line, the DEAN FOREST to MUSKOGEE 230.0 kV loads to 105.2% of its 861 MVA rating. Project DFAX: 9.3%.

MUSKOGEE 500-230 kV Transformer Bank: 105.0% loading (1,410 MVA) following the outage of the MUSKOGEE to SWAN VLY 500.0 kV; DFAX = 12.0%.

For the loss of SABANERA to LONETREE 500.0 kV, the MONTPELIER ELECTRIC GENERATING STATION to WABASH 500.0 kV Line loads to 101.8% of its 2,847 MVA rating. Project DFAX: 28.4%.

SPY RUN to INDUSTRIAL PARK 100.0 kV line: 101.6% loading (138 MVA) following the outage of the WAYNEDALE to STATE ST 230.0 kV Line; DFAX = 5.4%.

For the loss of MONTPELIER ELECTRIC GENERATING STA to WABASH 500.0 kV, the MONTPELIER ELECTRIC GENERATING STATION 500-230 kV Auto loads to 100.3% of its 1,479 MVA rating. Project DFAX: 16.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AEP PETTIT AVENUE to WAYNEDALE SUBSTATION 100.0 kV #1	Rebuild line with higher-capacity conductor	\$1,614,000	37% (\$602,000)
WAYNEDALE to AEP PETTIT AVENUE 100.0 kV	Rebuild line with higher-capacity conductor	\$1,727,000	37% (\$644,000)
INDEGO to WAYNEDALE 100.0 kV L	Rebuild line with higher-capacity conductor	\$4,360,000	34% (\$1,489,000)
WABASH to MONFORT SUBSTATION 230.0 kV	Rebuild line with higher-capacity conductor	\$27,446,000	21% (\$5,842,000)
INDEGO to COLUMBIA HILLS 100.0 kV	Reconductor line	\$1,039,000	34% (\$351,000)
PEAVINE to HOHENWALD 100.0 kV L	Reconductor line	\$646,000	24% (\$156,000)
MONFORT to NE HARRIMAN 100.0 kV line	Reconductor line	\$834,000	53% (\$439,000)
PONNEQUIN to DEER CREEK 100.0 kV L	Reconductor line	\$1,141,000	52% (\$594,000)
EASTOWN ROAD to DEAN FOREST 230.0 kV	Reconductor line	\$9,088,000	17% (\$1,528,000)
DEAN FOREST to MUSKOGEE 230.0 kV line	Reconductor line	\$14,392,000	17% (\$2,420,000)
MUSKOGEE 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$28,533,000	13% (\$3,769,000)
WABASH to MONTPELIER ELECTRIC GENERATING STA SUBSTATION 500.0 kV L	Reconductor line	\$40,926,000	14% (\$5,639,000)
SPY RUN to INDUSTRIAL PARK 100.0 kV	Reconductor line	\$750,000	18% (\$137,000)
MONTPELIER ELECTRIC GENERATING STATION 500-230 kV Auto	Replace transformer with higher-rated unit	\$43,988,000	9% (\$4,121,000)
	Total allocated to project		\$27,731,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.